

Customer Churn Prediction & Retention Analytics

Business case study • Using customer data and machine learning to identify churn risk early

1. Executive Summary

Customer churn directly affects recurring revenue and makes growth more expensive when businesses have to replace customers instead of retaining existing ones. This project uses the IBM Telco Customer Churn dataset to build a practical churn-risk workflow: clean the customer data, identify important churn patterns, compare machine-learning models, and convert predicted probabilities into customer risk segments.

Key outcome: a Tuned Random Forest was selected because it produced the strongest F1 score (63.68%) and recall (79.68%) among the final candidates. For a retention use case, recall matters because missing a likely churner can mean losing the opportunity to intervene. The model was then used to segment 7,043 customers into high, medium and low predicted-risk groups.

2. Data & Methodology

The dataset contains 7,043 customer records covering account information, services, billing characteristics and churn status. The target variable was **Churn Label**, encoded as 0 for No and 1 for Yes.

Stage	What was done
Data preparation	Checked structure, data types, missing values and duplicates; converted Total Charges to numeric.
Feature engineering	Created average monthly spend, tenure groups and a total-services count.
Preprocessing	Used imputation, IQR-based outlier capping, scaling for numerical variables and one-hot encoding for categorical variables.
Modeling	Compared Logistic Regression, Decision Tree, Random Forest and XGBoost, followed by tuning of Random Forest and XGBoost.
Evaluation	Compared accuracy, precision, recall, F1 and ROC-AUC; confusion matrix and classification report were also reviewed.

Leakage control: churn-derived fields such as Churn Score, Churn Value and Churn Reason were excluded from modeling. This keeps the prediction task focused on information that would be available when making a retention decision.

3. What the Data Tells Us

- **Contract type:** month-to-month customers are the clearest high-risk group compared with customers on longer contracts.
- **Tenure:** newer customers are more exposed to churn, while longer tenure generally acts as a protective signal.
- **Service mix:** fiber optic internet appears among the strongest model explanation signals.
- **Support:** customers without tech support or online security appear more exposed to churn risk.
- **Billing and price:** electronic check and higher monthly charges are among the important signals identified by the model.

4. Machine Learning Results

Six model configurations were compared. The business objective was not simply to maximize accuracy; the preferred model should identify a large share of potential churners while maintaining a reasonable balance between precision and recall.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	74.88%	51.75%	78.88%	62.50%	84.91%
Decision Tree	73.95%	50.60%	79.14%	61.73%	84.11%
Random Forest	76.79%	54.49%	76.20%	63.55%	85.32%
XGBoost	80.84%	66.77%	55.35%	60.53%	85.62%
Tuned Random Forest	75.87%	53.02%	79.68%	63.68%	85.20%
Tuned XGBoost	80.20%	65.37%	54.01%	59.15%	85.55%

Model selection: Tuned Random Forest was selected. XGBoost achieved higher accuracy and slightly higher ROC-AUC, but its recall was much lower. For a retention workflow, the tuned Random Forest provides the better trade-off because it catches more of the customers who are likely to churn.

5. Customer Risk Segmentation

Segment	Rule	Customers	Share
High	Probability $\geq$ 70%	1,557	22.1%
Medium	$40\% \leq$ Probability $<$ 70%	1,845	26.2%
Low	Probability $<$ 40%	3,641	51.7%

The risk score should be treated as a prioritization tool rather than a guarantee that a customer will churn. The main value is deciding where limited retention effort should be concentrated.

6. Key Churn Drivers & Retention Actions

Observed signal	Suggested business response
Month-to-month contract	Prioritize renewal conversations, longer-term plan options and proactive check-ins.
New / low-tenure customers	Strengthen onboarding, early-life engagement and first-month service follow-up.
No tech support / online security	Offer relevant support or service education before frustration becomes a churn event.
Fiber optic service	Review service experience, support issues and plan fit for high-risk customers.
Higher monthly charges	Test targeted plan recommendations or value-based offers rather than blanket discounts.
Electronic check	Review payment experience and encourage smoother recurring-payment options where appropriate.

7. Conclusion & Next Step

The analysis shows that churn is not evenly distributed across the customer base. Contract type, tenure, service mix, support and billing characteristics provide useful signals for identifying customers who may need attention. The Tuned Random Forest offers a practical balance for this retention-focused use case, with 79.68% recall and a 63.68% F1 score.

Recommended next step: connect the risk score to customer-level retention workflows and measure the actual outcome of interventions. The most important business test is not only whether the model predicts churn, but whether acting on those predictions reduces avoidable churn and improves retention efficiency.

Portfolio note: This analysis uses the IBM Telco Customer Churn public dataset.